

S.3 Geography

Topic 2: Living with Natural Hazards

Digital Textbook

Tectonic & Climatic Hazards

Introduction: Natural Hazards (Textbook P.4)

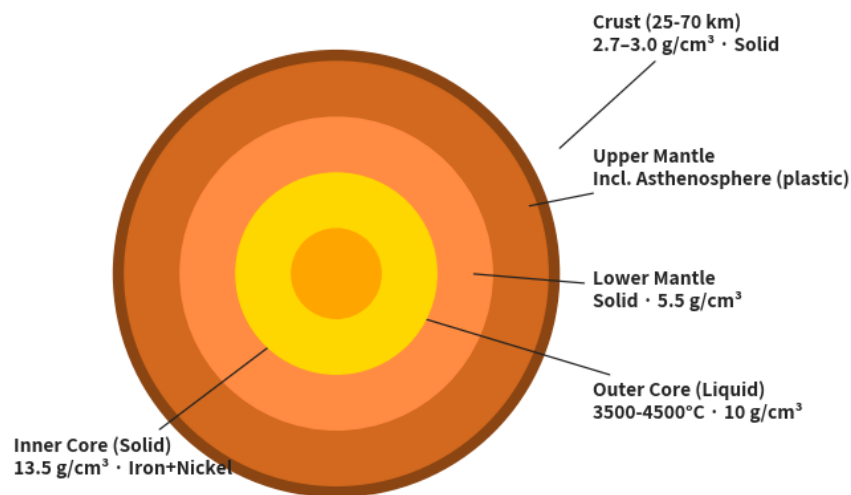
Type	Cause	Examples
Tectonic hazard	earth movement	(1) Earthquake (2) Volcanic eruption (3) Tsunami
Climatic hazard	extreme climate	flooding, drought, tropical cyclone (Typhoon)
Geological hazard	geological process	landslide, avalanche
Biological hazard	living organism	pests, disease, bacteria

Part I. Tectonic Hazards

1. Structure of the Earth (Textbook P.48)

- The Earth is made up of 3 layers: (1) Crust (2) Mantle (3) Core
- They are arranged according to their density, pressure, temperature

Structure of the Earth

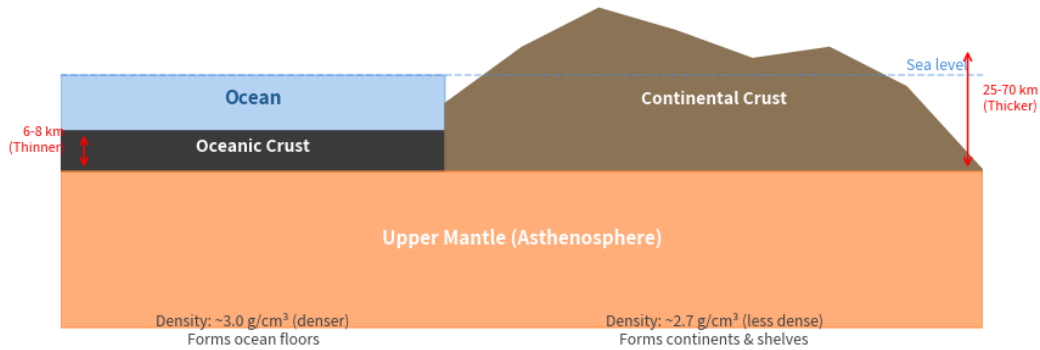


- (1) The crust is made up of oceanic crust and continental crust.
- (2) The mantle is made up of upper mantle and lower mantle.
- (3) The core is made up of outer core and inner core.

① Crust

The crust is made up of **continental** crust (*not covered by sea*) and **oceanic** crust (*always covered by seawater → forms ocean floors*). It is in solid state.

Comparison: Continental vs Oceanic Crust



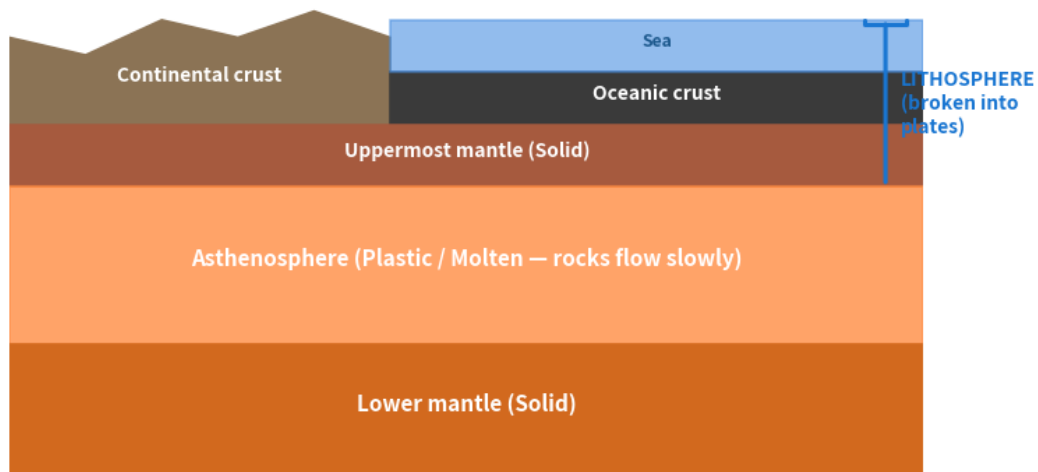
Comparison between continental crust and oceanic crust

	Continental crust	Oceanic crust
Thickness	<u>Thicker</u>	<u>Thinner</u>
Density	<u>Less, lower density</u>	<u>Denser, higher density</u>
Landform	forms continents and continental shelves	forms ocean floors

② Mantle

- The mantle is made up of **upper** mantle and **lower** mantle.
- **Upper mantle:**
 - consists of **the uppermost part of the mantle (solid state)** and **asthenosphere** (soft rocks) (plastic / molten state) *e.g. butter*
 - the uppermost part of the mantle and the crust are collectively known as the **lithosphere**
- **Lower mantle:** **Solid** state (*∵ they are both SOLID STATE*)

Lithosphere & Asthenosphere



③ Core

- The core is made up of **outer core** and **inner core**.
- **Outer core:** Liquid (*less dense*) state (high temperature melts metal into liquid)
- **Inner core:** Solid (*denser*) state (high pressure, material maintains solid)

(Additional): magma → cooldown → rocks → melting → magma (cycle)

Comparison between Lithosphere and Asthenosphere

Earth Layer	Sub-layer	Physical State	Lithosphere / Asthenosphere
Crust	Continental crust	Solid	LITHOSPHERE (solid) → broken into <i>plates</i>
	Oceanic crust (sea)	Solid	
Upper Mantle	Uppermost part of the upper mantle	Solid	
	Lower part of upper mantle (soft rocks that flow slowly)	Plastic / Molten	ASTHENOSPHERE (plastic / molten)
Lower Mantle	Lower mantle	Solid	Lower mantle (solid)
Core	Outer core	Liquid	Outer core (liquid)
	Inner core	Solid	Inner core (solid)

💡 **Key idea:** The lithosphere and asthenosphere are defined by **physical state** (solid vs plastic), not by chemical composition.

LITHOSPHERE

= Crust + **uppermost SOLID** part of upper mantle
 → broken into **tectonic plates**

ASTHENOSPHERE

= the **plastic / molten** part of upper mantle, below the lithosphere
 → plates **float and move** on it

MCQ-1: 2013-Q9

Refer to the figure below.

[Figure: Cross-section showing Layer X (Continental crust above sea level) and Layer Y (Oceanic crust under sea), both above Upper mantle]

Which of the following pairs of comparison between layer X and layer Y in the figure are correct?

		Layer X	Layer Y
(1)	Density	higher	lower
(2)	Thickness	thinner	thicker
(3)	Chemical composition	silica and magnesium	silica and aluminium

A. (1) and (2) only

B. (1) and (3) only

C. (2) and (3) only

D. **(1), (2) and (3)** ✓ *Answer*

2. Plate Tectonics (Textbook P.49–50)

① Tectonic Plates

- The lithosphere is cracked and broken into pieces called **Tectonic Plates**.

Characteristics of plates:

- ► irregular shapes
- ► vary in thickness (*continental: thicker > ocean*)
- ► have different sizes
- ► lie and float on the asthenosphere
- ► move slowly in different directions *~5-10 cm/year*

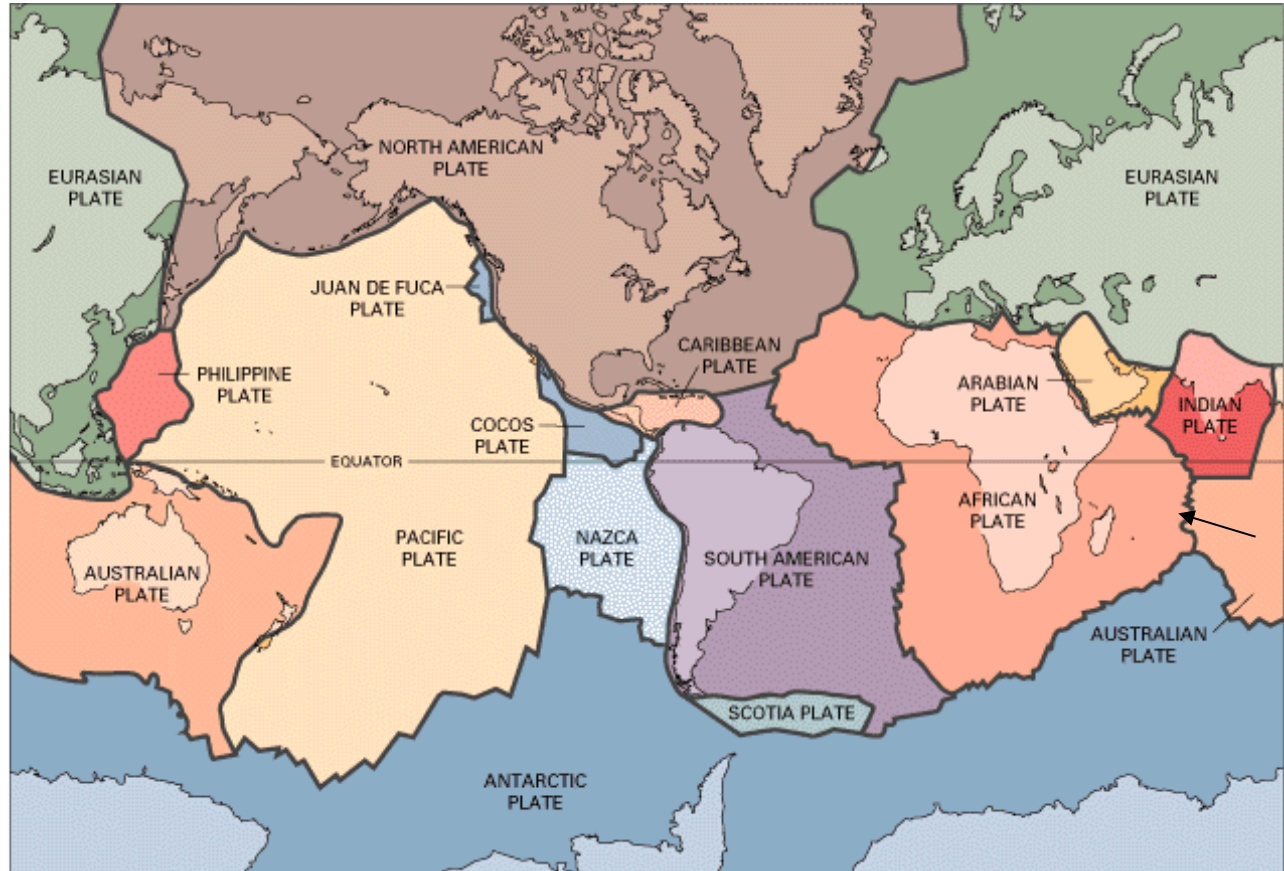
Plate A ↔ Plate B → apart → float on asthenosphere

The 15 Major Tectonic Plates

★ *Need to memorise: NAME, location, spelling* ★

No.	Plate name	Located near
1.	Eurasian Plate	Europe & Asia
2.	Arabian Plate	Arabian Peninsula
3.	African Plate	Africa
4.	Indo-Australian Plate	India & Australia
5.	Antarctic Plate	Antarctica
6.	Philippine Plate	Philippine Sea
7.	Okhotsk Plate	Sea of Okhotsk
8.	Pacific Plate	Pacific Ocean
9.	Juan de Fuca Plate	
10.	North American Plate	North America
11.	Cocos Plate	
12.	Caribbean Plate	Caribbean Sea

13.	Nazca Plate	<i>(common asked in exam)</i>
14.	South American Plate	South America
15.	Scotia Plate	

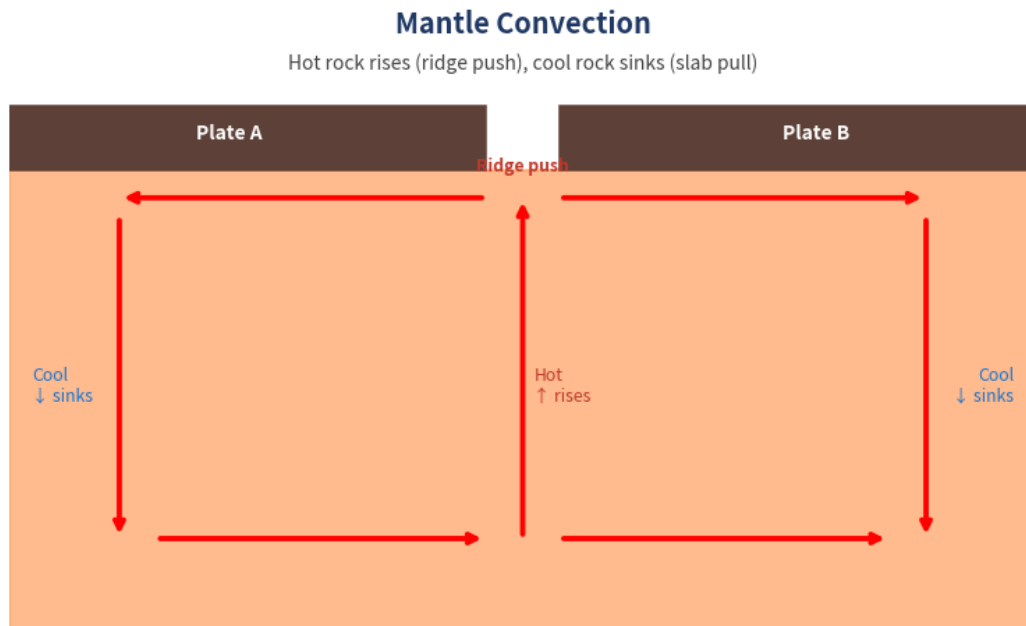


② Plate movement

- **Mantle Convection:**

- ➤ Rock in the lower mantle is heated, then expands and rises slowly
- ➤ Rock near the earth's surface is cooled, then contracts and sinks

- **(1) Mantle convection**, together with **(2) ridge pull** and **(3) slab pull**, drive the plates to move in different directions.



③ Plate boundary

- The border between two plates is called **plate boundary**

- There are 3 types of plate boundaries:

- (1) Convergent plate boundary (**Destructive** plate boundary) → *collide / move away*

- (2) Divergent plate boundary (**Constructive** plate boundary)

- (3) Transform plate boundary (**Conservative** plate boundary) → *slide past*

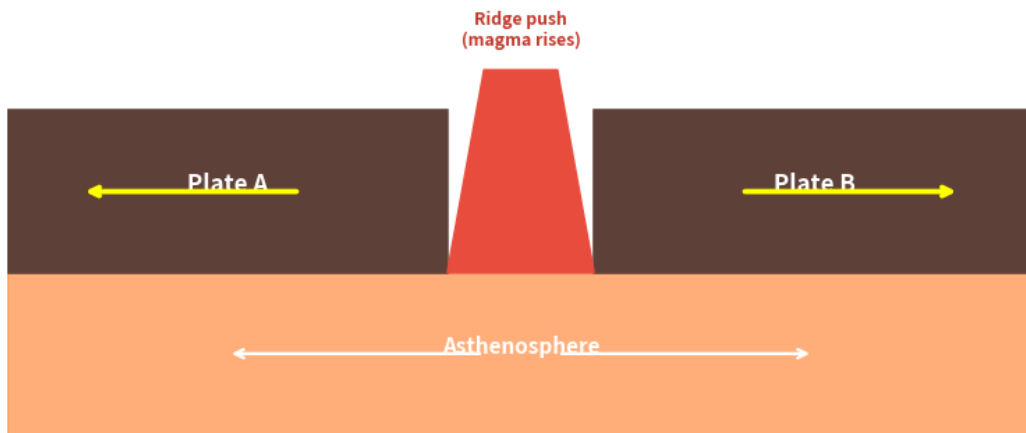
(Iceland: volcano → rocks cools down → magma)

S.3 Geography – C2 Natural Hazards

Comparison of 3 Plate Boundaries

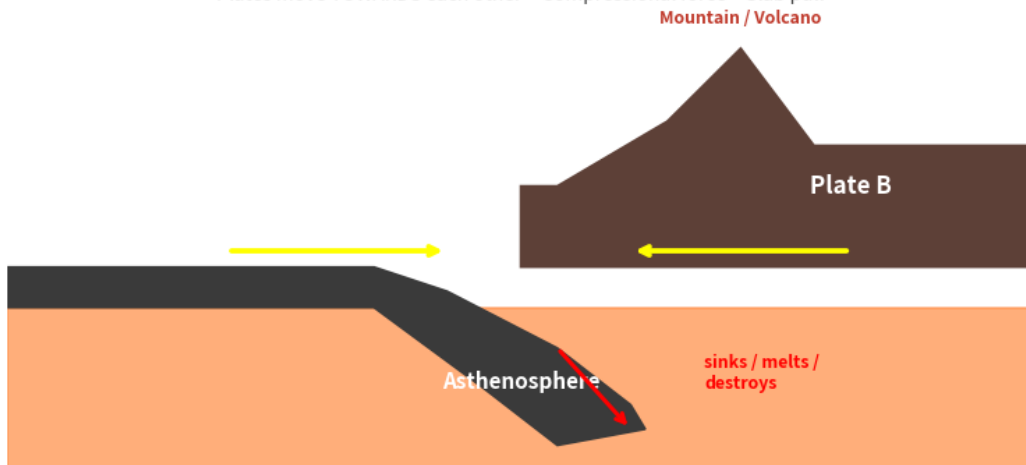
Divergent / Constructive Plate Boundary

Plates move AWAY from each other · Tensional force



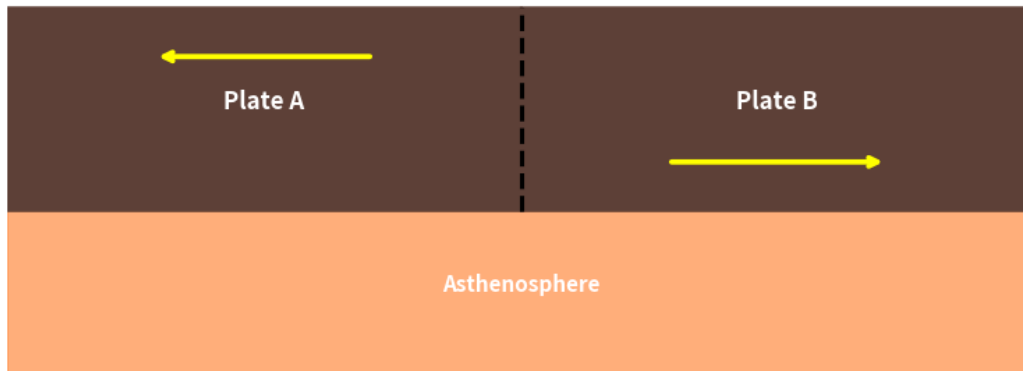
Convergent / Destructive Plate Boundary

Plates move TOWARDS each other · Compressional force · Slab pull



Transform / Conservative Plate Boundary

Plates SLIDE PAST each other horizontally · Shearing/lateral force



	(1) Divergent / Constructive	(2) Convergent / Destructive	(3) Transform / Conservative
Reason	Divergence of convection current in mantle; ridge push	Convergence of convection current in mantle; slab pull	Convection currents flow past each other
Plate movements	- plates diverge - move away from each other	- plates converge / collide - move towards each other	plates slide past each other horizontally
Internal Force	Tensional force (<i>Tension</i>)	Compressional force	Shearing / lateral force
Example	North American & Eurasian plate (Atlantic Ocean); Pacific & Nazca plate (Pacific Ocean)	Philippine & Eurasian plate; Indo-Australian & Eurasian plate	Pacific plate and North American plate

★ MEMORISE: Formation of Earthquakes ★

(depends on the location)

- Convergence** of convection current in mantle **drag** the X plate and the Y plate to **collide with each other** along the **convergent** plate boundary under **compressional force**.
- Divergence** of convection current in mantle **pull** the X plate and the Y to **move away from each other** along the **divergent** plate boundary under **tensional force**.
- Two convection current in mantle **flow past each other**, the X plate and the Y plate **slide past each other horizontally** along the **transform** plate boundary under **shearing force**.

3. Location of Plate Boundary

① Divergent Plate Boundary

No.	Zone	Description
(1)	<u>Mid-Atlantic</u>	north to south across the Atlantic Ocean
(2)	<u>Mid-Indian</u>	cutting the red sea and the Indian Ocean
(3)	Antarctic	forms a circle around Antarctica
(4)	<u>Southeast Pacific</u>	From Mexico to Antarctica

- Nearly all divergent plate boundaries are located under the ocean
- **Volcanoes** can be found (e.g. Iceland, Easter Island)

② Convergent Plate Boundary

No.	Belt	Description
(1)	<u>Alpine–Himalayan</u>	runs from the Mediterranean Sea, crosses the north of India and turns south-east into the Indian Ocean
(2)	<u>Circum-Pacific</u>	around the Pacific Ocean

(also known as the Pacific Ring of Fire 環太平洋火山帶 / 環太平洋地震帶)

- **Volcanoes** and **volcanic island arc** can be found (*e.g. Japan, Taiwan, Philippines, Indonesia*)

③ Transform Plate Boundary

The most famous transform plate boundary: The San Andreas Fault

- The San Andreas Fault is the most **active transform plate boundary**
 - North American plate and Pacific plate slide past each other
- Length: 200 km Location: California, USA
- Frequent earthquake occurs

4. Tectonic Hazard (Earthquake) (Textbook P.51–53)

- There are 3 tectonic hazards:
 - ► Earthquake
 - ► Volcanic eruption
 - ► Tsunami
- Most tectonic hazards are found along plate boundaries

1. Distribution of Earthquakes

- Most earthquakes occur at convergent and transform plate boundaries
- Fewer on the seabed at divergent plate boundaries
- They also occur at places away from plate boundaries (*rare*)
- Earthquake is the tectonic hazard affecting the largest area of the world

2. Formation of Earthquake

(Also apply when they ask: 'Why tsunami are common in that region?')

- **Earthquake** is the **sudden shaking** of rocks in the Earth's crust

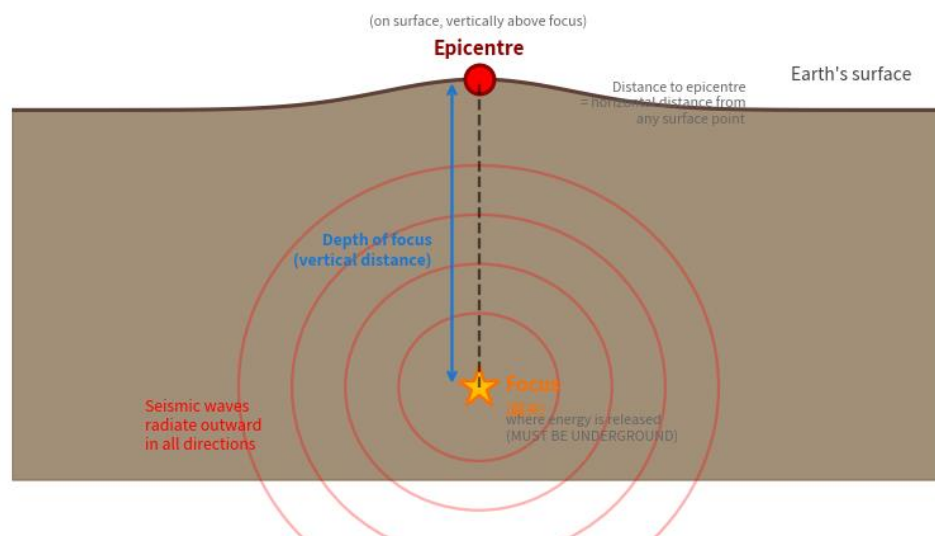
1.	Plate Tectonics (P.8)
2.	Stress applied to the rocks. Plates are held back by friction .
3.	Pressure builds up within the rocks and stress increases.
4.	When the stress exceeds the strength of the rock, the rock breaks and displaces . (<i>cannot push & start → break → displace</i>)
5.	The energy stored is released and transmits as seismic wave (地震波).
6.	The seismic waves move out in all directions and shake the ground , earthquake is formed .

Key terms

Focus (震央): the point where energy is released (*MUST BE UNDERGROUND, include the depth*)

Epicentre: the point on the earth's surface vertically above the focus

Earthquake: Focus and Epicentre



★ Near to Focus → Stronger feeling of earthquake

Examples of Places Affected by Earthquakes

Textbook Reference: Map showing locations along the convergent plate boundary (Pacific Ring of Fire) — Volcanic island arcs and ocean trenches.

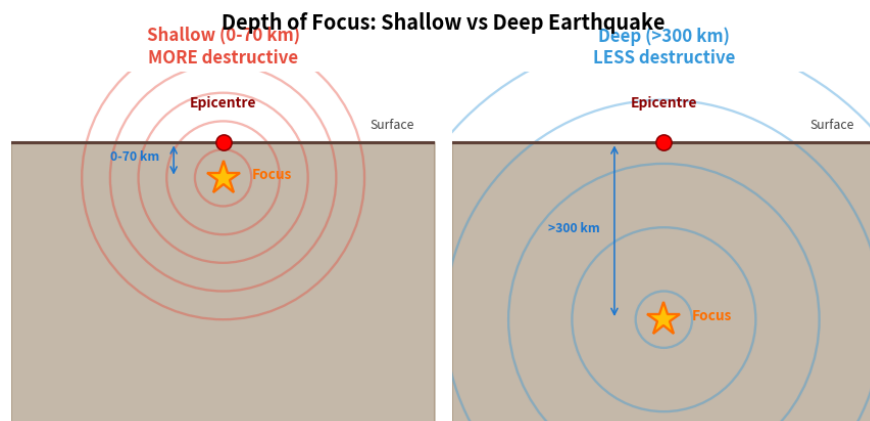
Example	Location	Plate 1	Plate 2
1	Japan	Okhotsk Plate	Pacific Plate
2	Ryukyu Island	Eurasian Plate	Philippine Plate
3	Philippines	Eurasian Plate	Philippine Plate
4	Indonesia	Eurasian Plate	Indo-Australian Plate
5	Marianas Islands	Philippine Plate	Pacific Plate
6	Aleutian Islands	North American Plate	Pacific Plate

3. Describing an Earthquake

There are 3 ways to describe an earthquake.

(1) Depth of focus (震源深度)

Type	Depth	
► Shallow	<u>0–70 km deep</u>	<u>No need to memorise the numbers</u>
► Intermediate	<u>70–300 km deep</u>	
► Deep	<u>>300 km deep</u>	



- Shallow earthquake is **more destructive**, as there is **less energy loss** when seismic waves transmit to the Earth's surface
- Which kind of earthquake is less destructive? Deep earthquake

(2) Intensity of earthquake

- Use modified Mercalli scale (修訂麥加利地震烈度表)
- Measure the **degree of damage** of a particular place (★ *different in different places*)

e.g. (X) 10 → *intense* (II) 2 → *weak* (XII / 12) → *cataclysmic*

(3) Magnitude of earthquake

- Use Richter scale (黎克特制地震震級)
- Measure the **amount of energy released** from the focus
- When the magnitude is increased by 1, the energy released is increased by **31.6 times**
(★ *Fixed in all places*)

e.g. M7.0 → M8.0 = *Difference: 31.6 times more energy released*

Richter Scale: Approximate Effects

Magnitude (R)	Effects
M1 – 2.9	Too weak to be felt by people but can be detected by seismometers.
M3 – 3.9	Felt by people on upper floors. Hanging objects swing.
M4 – 5.9	Felt by almost everyone. Buildings with poor structure collapse.
M6 – 8.9	Difficult for people to remain standing. Most buildings collapse. Roads crack.
M9 and above	Complete destruction. All buildings collapse.

5. Impacts of Earthquake (Textbook P.53–55)

Short Term Effects	Long Term Effects
• Destruction of <u>buildings and houses</u> • Destruction of communication and transportation network <i>e.g. roads, railway, bridges, base stations</i> • Hinder <u>rescue work</u> • Destroy public utilities <i>e.g. water, gas, electricity supplies</i> • Lead to <u>fire</u> (leakage of gas pipes) • <u>Tsunami</u> • Landslides / avalanches (崩) • Flooding (dam collapses) • Land subsidence (soil → liquid → no support → collapse) • Spread of <u>disease</u> (dead bodies cannot be removed quickly / shortage of clean water) • Aftershocks cause collapse of weakened structures 餘震 ↓ Economic loss, loss of life and property, people injured	• Obstruction of economic development • Bad effect on the psychology of the victims • Occurrence of riots and robbery (social instability)

Major Earthquakes – Data Table

Year	Location	Richter	Casualties	Major effects
1995	Kobe, Japan	7.2	Over 6,000	Structures damaged, fires
1999	Taiwan	7.6	Nearly 2,500	Structures damaged
2004	Off the west coast of northern	9.0	Over 280,000	Huge tsunami triggered and a volcano

	Sumatra, Indonesia			erupted
2008	Sichuan, China	7.9	Over 87,000	Structures collapsed, landslides
2011	North-east Japan	9.0	Over 20,000	Huge tsunami triggered
2015	Nepal	7.8	Nearly 9,000	Structures collapsed, landslides
2023	Turkey and Syria	7.8	Over 62,000	Structures collapsed
2024	Noto, Japan	7.5	Over 600	Structures collapsed, tsunami
2025	Myanmar and Thailand	7.7	Over 5,000	Structures collapsed
2025	Kamchatka, Russia	8.8	Less than 50	Structures collapsed, tsunami

6. Measures (Textbook P.56–59)

1. **Monitoring system and warning systems** *(to raise awareness and give time for early evacuation)*

- **Monitoring system:** equipment is installed to **detect the signs of natural hazards**
- **Warning system:** warnings are **issued through media** or **emergency alert system** when hazard strikes

Monitoring system

- Equipment to detect slight movements in the ground
- Seismograph is used to detect and measure seismic waves
- Scientists analyse past records of earthquakes

Warning system

- When an earthquake is detected, warnings are issued through **media** or **emergency alert system** (e.g. J-Alert in Japan)
- The information obtained is used to predict the **time and place of occurrence**, and the scale of possible earthquakes → evacuate or take safety precautions
- Example: Shinkansen in Japan conducts emergency stop when earthquake is detected

2. **Hazard-proof structures** *(to provide protection)*

A. For earthquake-prone regions

- Earthquake-proof design for buildings to prevent the collapse of structures
- Impose laws to ensure buildings can pass the earthquake-proof standard
- **Simple building materials** without earthquake-proof design is **prohibited**
- **Height is strictly controlled**, tall buildings are **designed to withstand strong earthquakes**

3. Land-use planning *(to avoid risky development)*

(Help government plan and formulate policies to ↓ impact)

- Produce 'hazard-zone map' according to level of risks
- Control development in different areas → reduce population in high-risk areas
 - → avoid building facilities with potential danger (e.g. nuclear power plants, oil)
- Establish evacuation routes with adequate road signs
- Large open areas (e.g. parks) are set up as safety evacuation areas
- School and indoor stadiums are designated as evacuation or shelter centres

4. Education and drills *(to increase public awareness, be prepared to evacuate)*

- In Japan, students are taught about the three tectonic hazards from primary school
 - → **increase public awareness** → the public knows how to react during a tectonic hazard
- Teach the public to pay attention to the emergency alert system
- Prepare **emergency kit**
- Set up education centres
- Regular large-scale drills → to raise public awareness

5. Rescue systems *(to reduce the number of deaths)*

- Injured people can have a higher chance to survive
- Well-trained rescue team can reduce the number of deaths
- Purchase advance equipment: life detectors, helicopters, medicines and wireless communication networks

6. Post hazard financial resources *(to rebuild the society after hazard)*

- **Disaster aid**
- **Insurance**

7. Level of Destruction Brought by Earthquakes (Textbook P.64–67)

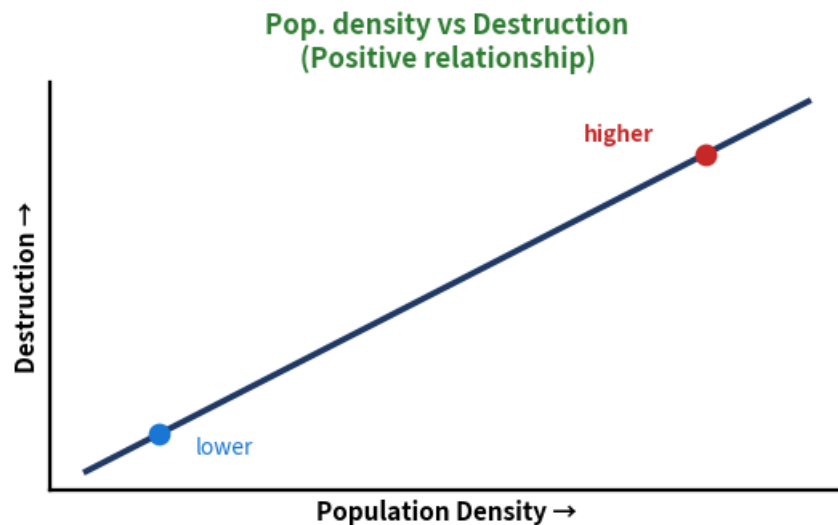
- The vulnerability of a place to tectonic hazards depends on 3 factors:

(1)	Characteristics of the affected area
(2)	Characteristics of the hazard
(3)	More Developed Countries and Less Developed Countries

(1) Characteristics of the Affected Area

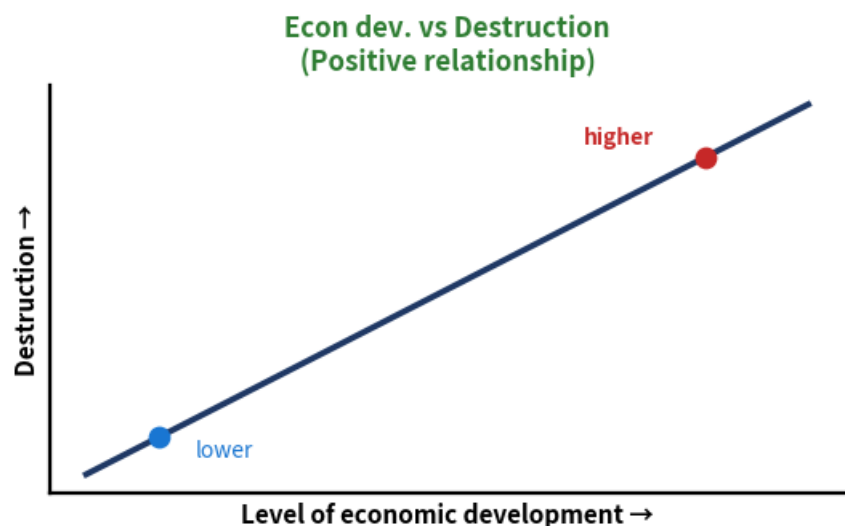
1. Population density of the area (city/town)

- The (lower / **higher**) the population density, the more the people exposed to the hazard



2. Level of economic development

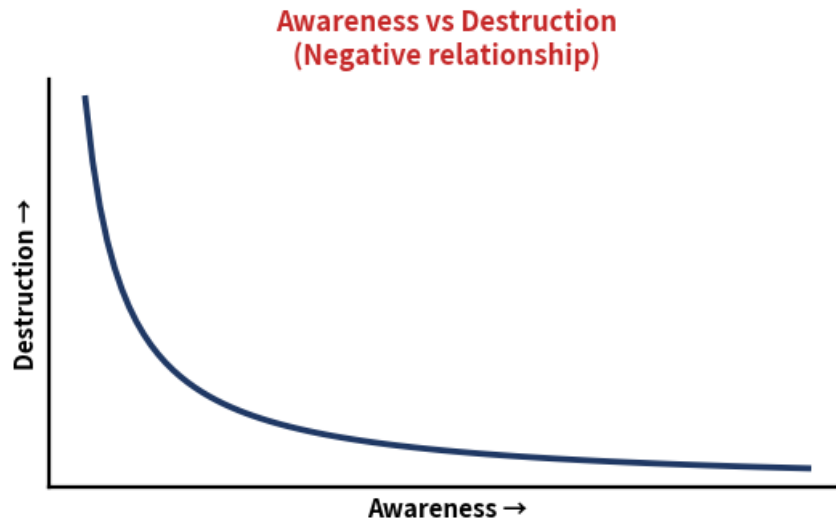
- **Higher level of economic development** → more economic infrastructure → **greater potential loss** of valuable **property**
 - e.g. during the night



3. Preparedness of people

◦ *vs. low awareness → more death and casualties*

- If people have **higher awareness** and are well-prepared for a hazard, they know **how to react** during the hazard. There is (greater / **less**) loss. e.g. evacuation



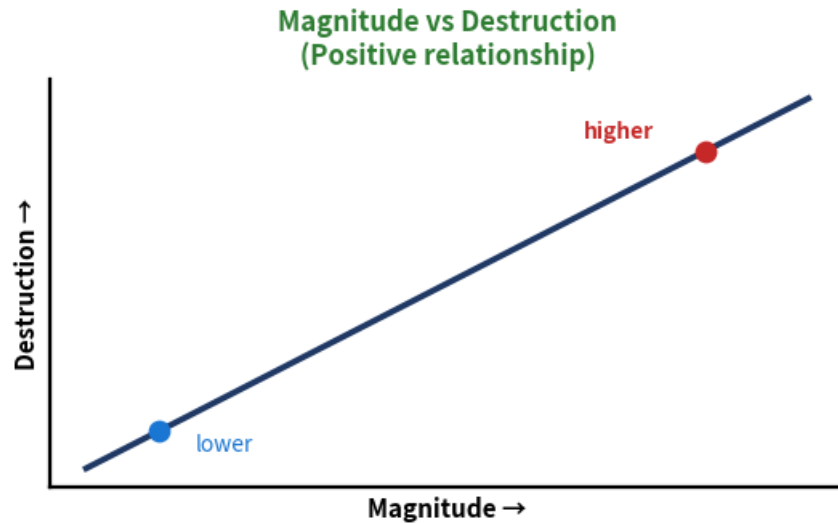
4. The weather after the hazard

- **Extreme weather** (such as cold surges and heat waves). **Increases** / Reduces death toll in areas where shelter and shade are insufficient.

(2) Characteristics of the Hazard

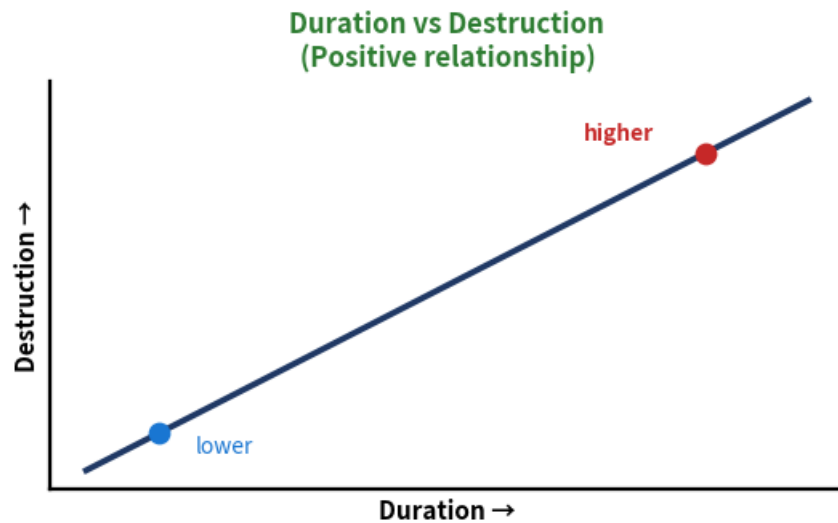
1. Magnitude

- Earthquake with a (lower / **higher**) magnitude, **more energy is released** and causes greater destruction.



2. Duration

- The **longer the duration**, the lower the ability of people and structures to withstand the hazard, results in **greater losses**.

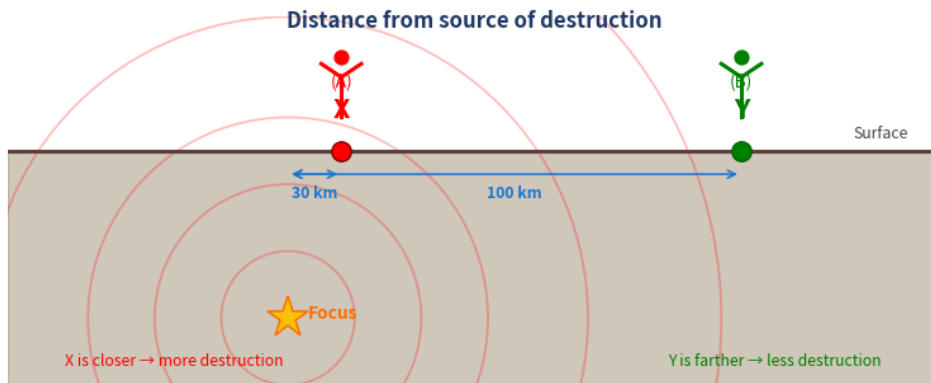


3. Time of occurrence

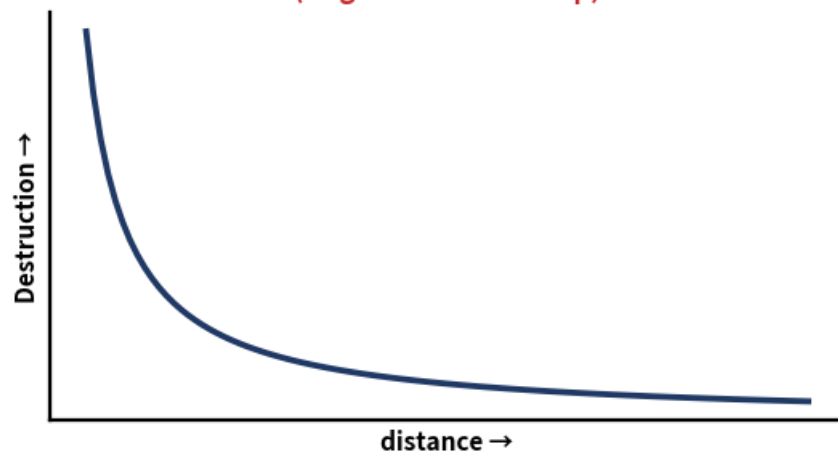
- *USE 'evacuate' instead of 'running away'*
- Destruction will be greater when people are not able to evacuate.
- e.g. **at midnight** (people are sleeping → **low awareness**) — *Need to say 'because of Sleeping → low Awareness'*

4. Distance from the source of destruction

- The **longer** the **distance** from the source of destruction, the **lower** the **destruction**.
- **Energy of seismic waves** (increases / **decreases**) as it travels. Therefore, there will be more destruction when:
 - **Distance** to the epicentre (horizontal distance) is **short**
 - **Depth of focus** (vertical distance) is **shallow**



Distance vs Destruction (Negative relationship)



5. Frequency of the hazards

- People tend to be **less prepared** to take precautions when there are **less frequent hazards**.

6. Height of tsunami wave

- The higher the tsunami wave, the more energy is released, causing greater destruction.

(3) Level of Economic Development (MDCs and LDCs)

(refer to Classwork WS in Classwork Notes)

- In general, the LDCs suffer more from a hazard

1. Level of economic development

(e.g. Afghanistan)

- LDCs lack capital and technology to cope with tectonic hazards
- LDCs are not able to conduct research, develop warning system and improve rescue equipment
- People did not receive warnings and did not evacuate, and government cannot afford rescue work

2. Transport network

- In LDCs, the transport network is often poorly developed → rescue work is hindered (not efficient) when road is blocked by landslide

3. Strength of buildings

- Many buildings in LDCs are built using simple building materials (e.g. stones, bricks and wood), which cannot withstand heavy shaking and fire
- Lack of earthquake-proof designs and building standards

4. Population density

- The population density is usually (lower / higher) in cities in LDCs, leading to higher casualties

5. Social value, education and people's awareness

- People with poor education level are (more / less) prepared for hazards ↗ *religious beliefs*
- People with strong traditional values and old-fashioned ideas, which is less likely to change the way they live and participate in drills and evacuation plans

6. Government efficiency

- Governments of LDCs may not respond quickly enough to help the victims
- Corruption prevents resources from reaching the affected area
- Lack of law enforcement on building codes and earthquake-proof designs

Exercises: Past Paper MCQs

MCQ-2: 2014-Q9

Which of the following statements about the structure of the earth are correct?

- (1) The asthenosphere is thicker than the earth's crust
- (2) The lower mantle is in molten form. **X** *(Incorrect — lower mantle is solid)*
- (3) The outer core is less dense than the inner one.

- A. (1) and (2) only
- B. (1) and (3) only ✓ *Answer*
- C. (2) and (3) only
- D. (1), (2) and (3)

MCQ-3: 2016-Q11

There were only 13 deaths in the 2015 earthquake in a place which measured 8.3 on the Richter Scale. Which of the following best explain the low death toll in the earthquake?

- (1) Government issued preventive warnings. ✓
- (2) The earthquake mainly affected less densely populated regions. ✓
- (3) Weak building materials in the rural areas.

- A. (1) and (2) only ✓ *Answer*
- B. (1) and (3) only
- C. (2) and (3) only
- D. (1), (2) and (3)

MCQ-4: 2017-Q12

Which of the following measures may help to reduce casualties in earthquake?

- (1) earthquake-proof structures ✓
- (2) well-equipped rescue teams ✓

(3) earthquake drills

A. (1) and (2) only

B. (1) and (3) only

C. (2) and (3) only

D. (1), (2) and (3) ✓ *Answer*

Part II. Climatic Hazards

1. Introduction to Weather and Climate (Textbook P.8)

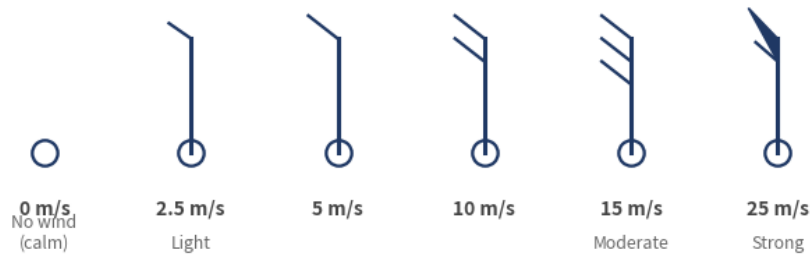
- **Weather:** conditions of the lowest part of the atmosphere over a **short period of time**.
- **Climate:** average weather conditions of a place over a **long period of time**, e.g. 30 years.

Key elements of weather and climate:

Element	Unit	Element	Unit
Wind speed	<u>m/s, km/h</u> ($m/s \times 3.6 = km/h$)	Temperature	<u>°C / °F</u> (water/alcohol)
Wind direction	<u>Compass point</u> (N, NE, E, etc.)	Relative humidity	<u>%</u>
Rainfall	<u>mm</u> (30 mm/h / 50 mm/h Amber / Red warning signal)	Air Pressure	<u>hPa (hectopascal)</u>
Sunshine	<u>hour</u>		

2. Weather Chart (Textbook P.9)

Wind Speed Symbols (Meteorological)



Hint: Convert km/h → m/s by dividing by 3.6

Common Weather Map Symbols (Legends)

Legends

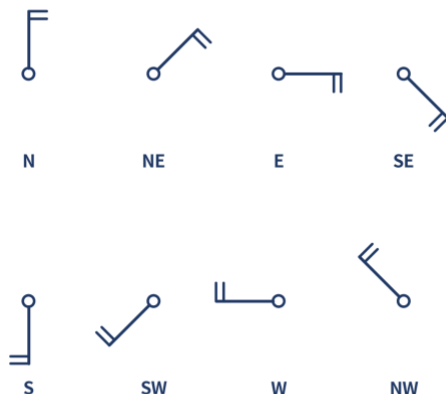
Drizzle	Haze	25m/s wind	Dissipating Cold Front
Rain	Ice/Snow/mixed with rain	Isobar (air pressure values in hPa)	Axis of Trough
Shower	Sand/Dust	Cold Front	Centre of Tropical Cyclone
Thunderstorm	Sandstorm/Duststorm	Warm Front	
Mist	Calm or light variable wind	Occlusion	
Fog	5m/s wind	Stationary Front	

Haze:	Caused by air pollutants	Shower:	Not long-lasting rain	Nothing can be found:	Nil
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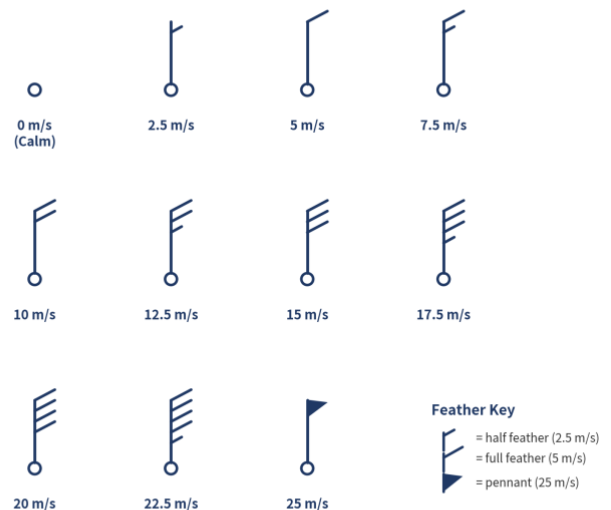
Isobar (氣壓線)

- Lines on a weather map joining points of equal **air pressure (in hPa)**

Wind Direction Symbols (8 Compass Directions)



Wind Speed Symbols — 0 to 25 m/s



★ The shaft points TOWARDS the direction the wind is coming FROM.
e.g. "N" = wind blowing FROM the north (shaft points up)

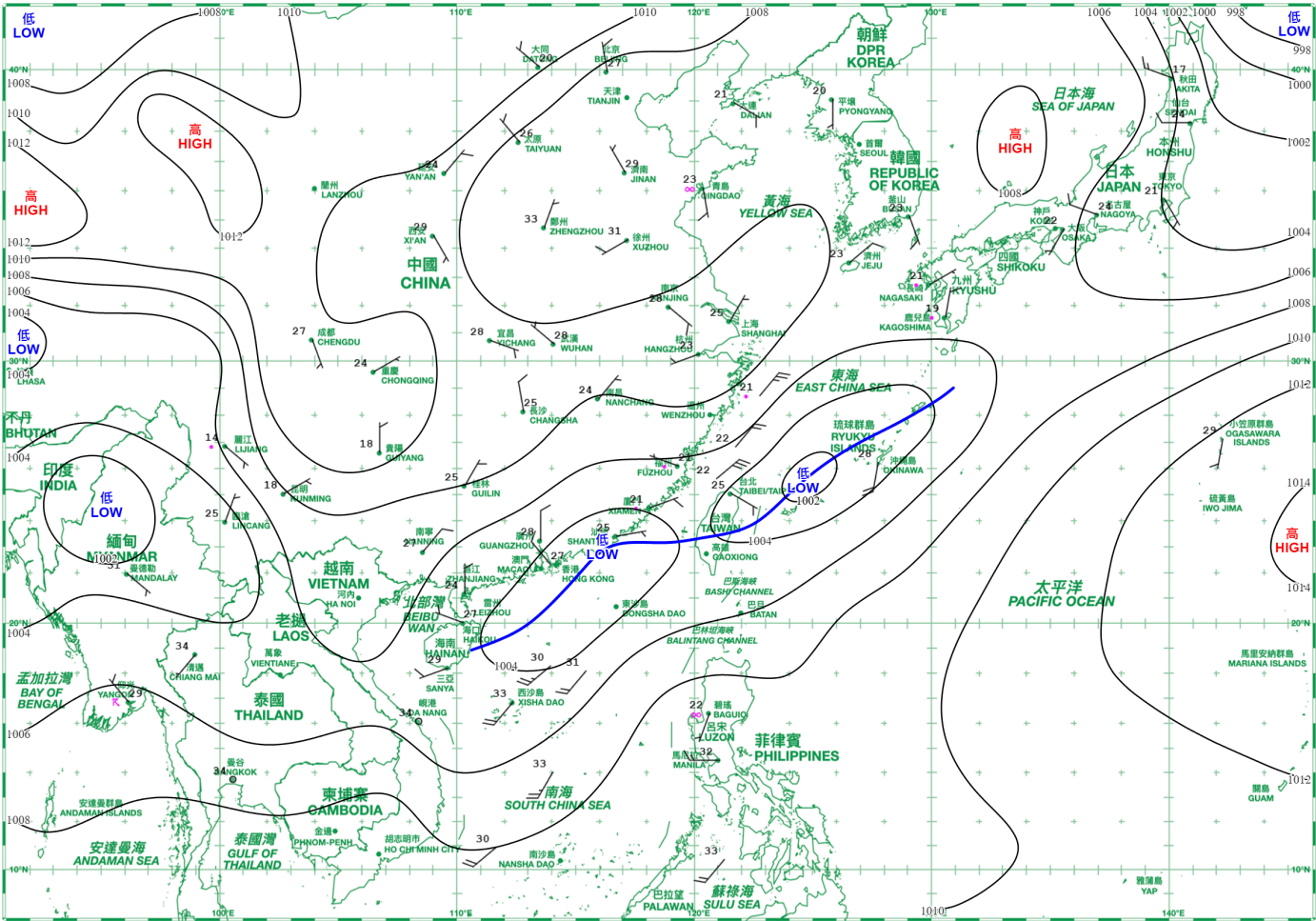
★ Always read from TIP of shaft downwards. Build the speed by ADDING feathers.
e.g. 7.5 m/s = 1 full feather (5) + 1 half feather (2.5)

Observation from Weather Chart (5 points)

Legends			
Drizzle	Haze	25m/s wind	Dissipating Cold Front
Rain	Ice/Snow/mixed with rain	Isobar (air pressure values in hPa)	Axis of Trough
Shower	Sand/Dust	Cold Front	Centre of Tropical Cyclone
Thunderstorm	Sandstorm/Duststorm	Warm Front	
Mist	Calm or light variable wind	Occlusion	
Fog	5m/s wind	Stationary Front	

City	Temperature (°C)	Wind Speed (m/s)	Wind Direction	Precipitation	Air Pressure (hPa)
Hong Kong	27 °C	5 m/s	Northwest	Nil	1004 hPa
Seoul	23 °C	2.5 m/s	Southeast	Nil	1006–1008 hPa
Tokyo	21 °C	5 m/s	Southeast	Nil	1004 hPa
Shanghai	25 °C	2.5 m/s	Northeast	Nil	1008–1010 hPa

日期/Date: 09.06.2026 香港時間/HK Time: 14:00 香港天文台 Hong Kong Observatory

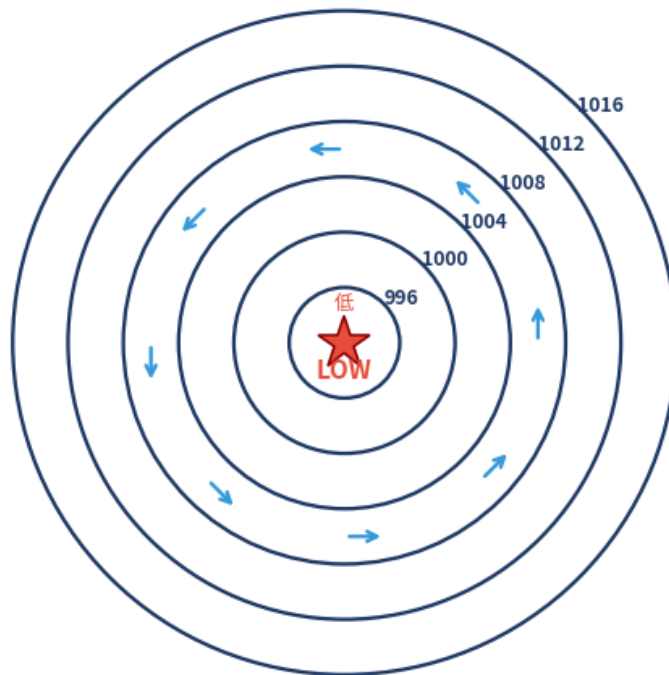


4. Tropical Cyclone (Textbook P.12–21)

1. Introduction

- **Tropical cyclone**, also known as **Typhoon** (*Hurricane* in North America)

Closely-packed isobars → steep pressure gradient → strong cyclone



Tropical Cyclone (Low-Pressure System)